Cryptocurrency Price-Direction Prediction

LSTM models with historical price data and media sentiment

Project	CSIT998 Group 16 capstone project
Version	1.0 - Revised requirements draft
Status	For team review and approval
Date	September 2026
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Approved project direction. The system will test whether adding media sentiment improves next-day cryptocurrency price-direction prediction relative to price-only models. Improvement is an empirical outcome to be measured, not a guaranteed result.

1. Purpose and success criteria

This specification defines the minimum functional and non-functional requirements for a reproducible research system that predicts whether the next UTC day's closing price will be higher or not higher than the current UTC day's closing price.

ID	Success criterion
SC-01	The complete BTC pipeline runs from approved raw data to final test predictions.
SC-02	Price-only and price-plus-sentiment LSTM models are evaluated fairly on the same untouched test dates.
SC-03	Results quantify whether sentiment changes performance and by how much, without requiring an improvement.
SC-04	A second team member can reproduce the main evaluation table from the repository instructions.
SC-05	A lightweight demonstration presents stored results clearly and is labelled as experimental, not financial advice.

2. Scope and fixed decisions

Decision	Approved position
Primary asset	BTC is mandatory. ETH is a Should-level extension using the same pipeline.
Prediction frequency	One prediction per asset per UTC day.
Primary target	Next-day direction: 1 if Close(t+1) > Close(t); otherwise 0.
Study period	Target January 2021-May 2025, adjusted only to the verified overlap of price and sentiment data.
Price source	Binance daily BTCUSDT klines; ETHUSDT for the extension.
Core sentiment source	An approved structured historical cryptocurrency news dataset with publication timestamps.
Optional sentiment source	Reddit only if authorised, timestamped historical data is obtained without delaying the core pipeline.
Excluded from core	Twitter/X scraping, live retraining, financial trading and automated trading advice.
Core models	Naive baseline, XGBoost baseline, price-only LSTM and price-plus-sentiment LSTM.
Sentiment method	VADER is mandatory; FinBERT is optional after the VADER experiment is complete.

3. Priorities and accountable roles

Priority	Meaning
Must	Required for the minimum acceptable project.
Should	Important and planned, but may be deferred if it threatens Must requirements.
Could	Optional extension attempted only after all Must requirements pass.

Code	Role	Primary member
DL	Data and governance lead	Pius Kalambe
PL	Pipeline and feature-engineering lead	Md Aminul Haque
ML	Modelling and evaluation lead	Almas Ashrafi
PC	Project coordination and reporting lead	Nguyen Gia Long
ALL	Peer review, testing and final approval	All team members

4. Data acquisition and governance requirements

ID	Priority	Requirement	Acceptance criterion	Owner
DR-01	Must	Collect daily BTCUSDT OHLCV records from Binance for the approved study period.	Each retained UTC date has one BTC row containing timestamp, open, high, low, close and volume.	DL
DR-02	Should	Collect ETHUSDT through the same pipeline after BTC data passes quality checks.	ETH records use the same schema, timezone and validation rules as BTC.	DL
DR-03	Must	Preserve raw source data separately from processed data.	Raw files are read-only inputs; all changes are produced by versioned scripts into processed folders.	DL
DR-04	Must	Record provenance for every dataset.	Source, owner, URL, retrieval date, coverage, licence/terms, row count and known limitations are documented.	DL
DR-05	Must	Use an approved historical news dataset as the core sentiment source.	Records contain publication timestamp, text or headline, source reference and sufficient overlap with the price period.	DL
DR-06	Could	Add Reddit only after access and coverage checks pass.	Permission/terms, timestamps, date overlap and collection method are documented; failure does not block the core project.	DL

5. Data quality and preprocessing requirements

ID	Priority	Requirement	Acceptance criterion	Owner
DQ-01	Must	Standardise all timestamps to UTC.	Processed timestamps are timezone-aware or explicitly documented as UTC; invalid timestamps are rejected and logged.	PL
DQ-02	Must	Remove duplicate asset-date price rows and duplicate news URLs/headlines.	No duplicate asset-date keys remain; text deduplication counts are reported.	PL
DQ-03	Must	Validate numeric and logical OHLCV values.	OHLCV fields are numeric; volume is non-negative; low is not above open/close/high and high is not below open/close/low.	PL
DQ-04	Must	Identify missing dates and values without silently fabricating observations.	A quality report lists missingness and the approved treatment; any imputation is documented and fitted without test leakage.	PL

ID	Priority	Requirement	Acceptance criterion	Owner
DQ-05	Must	Verify extreme price movements before correction.	Genuine market movements are retained; only confirmed source/data errors are corrected, with an audit record.	DL
DQ-06	Must	Classify text by asset relevance.	Each retained item is labelled BTC, ETH, both or irrelevant; irrelevant items are excluded from modelling.	PL

6. Sentiment and feature-engineering requirements

ID	Priority	Requirement	Acceptance criterion	Owner
SF-01	Must	Calculate a VADER compound score for each eligible text item.	Every retained item has a reproducible numeric sentiment score and source timestamp.	PL
SF-02	Must	Aggregate eligible sentiment by UTC date and asset.	Daily features include mean score, positive count, negative count and article count; zero-article days are explicitly handled.	PL
SF-03	Could	Repeat the sentiment experiment with FinBERT.	FinBERT uses the same items, dates, aggregation rules and test period as VADER.	PL
PF-01	Must	Generate causal price features from current and past observations only.	Features include close, volume, daily return, lagged close, moving averages and momentum; no feature uses future rows.	PL
PF-02	Must	Fit scaling and preprocessing parameters on training data only.	The saved training scaler is applied unchanged to validation and test data.	PL
PF-03	Should	Document sequence length and indicator windows.	Chosen windows are justified and identical across compared LSTM models unless explicitly tested as a controlled experiment.	PL

7. Temporal alignment and dataset preparation

Prediction cutoff: Only price and sentiment information available by 23:59:59 UTC on day t may be used to predict day t+1. Engagement measures that accumulated after the cutoff must not be used.

ID	Priority	Requirement	Acceptance criterion	Owner
TP-01	Must	Align price and sentiment using the defined UTC prediction cutoff.	Automated checks confirm that every input timestamp is on or before the associated prediction cutoff.	PL
TP-02	Must	Create the primary direction target from the next day's close.	Target equals 1 when Close(t+1) > Close(t), otherwise 0; the final row without a future close is excluded.	PL
TP-03	Should	Use day-t sentiment to predict day t+1 as the primary lag.	The primary model has a fixed one-day lag; alternative lags are separate, clearly labelled experiments.	PL
DS-01	Must	Produce one versioned modelling table.	The table contains date, asset, price features, technical indicators, sentiment features and target, with a data dictionary.	PL
DS-02	Must	Split data chronologically into training, validation and test periods.	The split is documented before final modelling; all models use identical test dates.	ML
DS-03	Must	Keep the test set untouched until the model design is frozen.	No test performance is used for feature selection, threshold choice, scaling, tuning or early stopping.	ML

8. Model requirements

ID	Priority	Requirement	Acceptance criterion	Owner
ML-01	Must	Implement a naive direction baseline.	The baseline predicts the majority training class and produces predictions for every test date.	ML
ML-02	Must	Implement an XGBoost price-feature baseline.	Training, validation and test dates match the LSTM experiment; settings and random seed are saved.	ML
ML-03	Must	Train Model A: price-only LSTM.	Model A uses only approved price/technical features and produces probabilities and classes for every test date.	ML
ML-04	Must	Train Model B: price-plus-sentiment LSTM.	Model B uses Model A's price features plus approved sentiment features and produces results on the same test dates.	ML
ML-05	Must	Control the comparison between Models A and B.	Models share sequence length, architecture, data periods, optimiser and training budget; any difference is documented and justified.	ML
ML-06	Must	Use validation data for model selection and early stopping.	Hyperparameters are chosen without consulting the test metrics; final settings are recorded.	ML

9. Evaluation and explainability requirements

ID	Priority	Requirement	Acceptance criterion	Owner
EV-01	Must	Evaluate all direction models on the same untouched test observations.	Report accuracy, balanced accuracy, precision, recall, F1-score and a confusion matrix.	ML
EV-02	Must	Produce a model comparison table.	The table reports each metric for the naive baseline, XGBoost, Model A and Model B.	ML
EV-03	Must	Quantify the sentiment contribution.	Report absolute metric differences between Models B and A and state whether results improved, worsened or were unchanged.	ML
EV-04	Must	Report variability and limitations honestly.	Random seeds are fixed and reported; major data, class-balance, coverage and generalisability limitations are discussed.	ML
EX-01	Must	Conduct basic feature and lag analysis without causal claims.	Report descriptive relationships between sentiment and subsequent price direction; correlation is not described as causation.	ALL
EX-02	Could	Use SHAP for an appropriate supported model.	SHAP outputs identify the analysed model and dataset and are treated as explanations, not causal effects.	ML

10. Demonstration interface requirements

ID	Priority	Requirement	Acceptance criterion	Owner
UI-01	Must	Provide a lightweight interface using saved data, predictions and metrics.	The demo loads without retraining and displays a clear data period and model version.	PC
UI-02	Must	Display historical price and sentiment trends.	Charts have labelled axes, units, asset and date range; missing sentiment days are not concealed.	PC
UI-03	Must	Display actual versus predicted direction and model metrics.	Users can compare the price-only and price-plus-sentiment models on the same test period.	PC
UI-04	Should	Allow selection between available assets.	BTC is always available; ETH appears only if its full pipeline and evaluation pass.	PC

ID	Priority	Requirement	Acceptance criterion	Owner
UI-05	Must	Display a research-use disclaimer.	The interface states that outputs are experimental and are not financial advice.	PC
UI-06	Could	Display a live current price.	If implemented, the source and retrieval time are displayed and failure does not block historical results.	PC

11. Software, repository and reproducibility requirements

ID	Priority	Requirement	Acceptance criterion	Owner
SW-01	Must	Maintain one shared Git repository with an agreed structure.	Repository contains data/raw references, data/processed, src, models, results and docs folders or documented equivalents.	ALL
SW-02	Must	Provide environment and execution instructions.	README and requirements.txt identify setup, commands, expected inputs and generated outputs.	ALL
SW-03	Must	Separate collection, cleaning, feature, training and evaluation code.	Each stage can be run independently and reports useful errors for missing files or columns.	PL
SW-04	Must	Protect credentials and restricted data.	API keys use environment variables; no credential or prohibited raw data is committed.	DL
SW-05	Must	Save model settings and outputs.	Random seed, features, split dates, hyperparameters, predictions, metrics and model/version identifiers are retained.	ML
SW-06	Must	Conduct an independent reproducibility check.	A member who did not build the final model reproduces the main evaluation table from the README and approved processed data.	ALL

12. Non-functional requirements

ID	Priority	Requirement	Acceptance criterion	Owner
NF-01	Must	Reliability: fail clearly on invalid inputs or unavailable dependencies.	Errors identify the failed stage and do not silently create incomplete outputs.	ALL
NF-02	Must	Reproducibility: preserve versions of data, code and configuration used for final results.	The report references an identifiable repository commit and modelling-dataset version.	ALL
NF-03	Must	Ethics and compliance: document source terms, privacy and redistribution limits.	The report explains how social/news data were obtained and what data can or cannot be shared.	DL
NF-04	Should	Usability: present results for a non-technical reader.	Labels avoid unexplained abbreviations and charts include concise interpretation notes.	PC
NF-05	Should	Performance: load the demonstration from saved artefacts in a practical time.	Normal demonstration use does not retrain models or recollect the full dataset.	PC

13. Required delivery package

Package	Minimum contents
Data	Raw-data references; processed modelling dataset; data dictionary; data-source register; quality and limitations report.
Code	Collection, preprocessing, sentiment, feature-engineering, training, evaluation and demonstration code.
Models	Saved final model artefacts and settings where feasible.
Results	Predictions, comparison metrics, confusion matrices and approved visualisations.
Documentation	README, dependency specification, methodology decisions, ethics/licensing notes and final technical report.
Demonstration	Lightweight interface using saved results and a research-use disclaimer.

14. Approval checklist

Approval item	Status
Prediction target fixed as next-day direction	[ ]
BTC confirmed as mandatory; ETH treated as extension	[ ]
Core price and sentiment sources approved	[ ]
UTC cutoff and leakage controls approved	[ ]
Owners and Must/Should/Could priorities approved	[ ]
Test period frozen before final evaluation	[ ]
All Must requirements passed or formally waived	[ ]

Team approval: _____ Date: _____